

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1708

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I M.Kavya/19247258 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Kavya M

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20

B

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

CO1 - Assessment Tool - 3

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CSA1708

Artificial Intelligence

Descriptive Written Test

i) An Intelligent agent is a system that perceives its environment through sensors & performs actions using actuators to achieve a goal. In a smart home, the agent controls lightning, temperature and security based on user behaviour & environmental conditions.

i) Simple Reflex agent:

- Works only on the current input
- Uses predefined condition-action rules
- Does not remember previous states.

For example: If motion is detected, it turns ON the light, if no motion is detected, it turns them OFF.

ii) Model-based Agent:

It keeps an internal memory of previous states.

It remembers past information and use it along with the current input to make better decisions.

For example: it remembers that the user usually enters the room at 7pm and prepares the room accordingly.

iii) Goal based agent:

It selects actions that help achieve a specific goal. Instead of simply reacting, it plan the best sequence of actions. For example, its goal may be to maintain a comfortable Temperature while reducing electricity consumption.

(iv) A utility Based Agent:

It chooses the action that gives the $\max^m$ overall benefit. It balances multiple factors such as comfort, energy efficiency & security. For example, it may slightly reduce air-conditioner usage to save electricity while still keeping the room comfortable.

Among all these, a hybrid approach combining Model-based & utility based component selects the best action by comfort. Therefore, its hybrid approach provides better performance in a smart home automation system.

2) A robot placed in an unknown maze must search for the exit without knowing maze prior knowledge.

i) Uniform Cost Search (UCS):

Uniform cost search expands the node with the lowest cumulative path cost.

Working:

- 1) start from the initial position
- 2) Expand the least-cost path
- 3) Continue until the goal is reached.

Advantages:

- Finds the optimal path
- Complete if steps costs are positive

Disadvantages:

- High memory usage
- Slow for large search space

Time complexity : $O(b^{\lceil (1+c^*/\epsilon) \rceil})$

Space complexity : $O(b^{\lceil (1+c^*/\epsilon) \rceil})$

where: b = branching factor

c^* = optimal path cost

ϵ = minimum step cost

ii) Iterative Deepening Search (IDS):

IDS repeatedly performs Depth-limited search with increasing depth limits until the goal is found.

Working:

- Search depth 0
- Then depth 1
- Then depth 2
- Continue until the exit is found

Advantages:

- Uses very little memory
- Complete
- Suitable for unknown-depth problem

Disadvantages:

- Repeats node expansion
 - Slower due to repeated searches
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Relation to Intelligent Agents:

- Simple reflex agent: cannot solve maze because only current input
- Model-based: Remembers visited locations and avoids revisiting paths.
- Goal-based: searches systematically to reach the exit.
- Utility-based: chooses the safest or least-cost path if multiple exits are available.

Conclusion:

The best combination is a Goal-based Agent with Uniform Cost search (UCS).

Reason:

- The goal-based agent focuses on reaching the exit
 - UCS guarantees the lowest cost & optimal path.
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RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand